

C267

Climate Change

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A science teacher at a school that installed rooftop solar panels last year explains to students why added carbon dioxide warms Earth. Which explanation is correct?

- A. CO₂ thins the ozone layer, letting in more sunlight.
- B. CO₂ absorbs incoming visible sunlight before it reaches the ground.
- C. CO₂ absorbs infrared energy given off by the surface and re-emits part of it back downward.
- D. CO₂ forms a solid layer that physically blocks air movement, like a glass roof.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

If Earth had no greenhouse gases at all, what would its average surface temperature be, compared with about 15 °C today?

- A. About –18 °C
- B. Exactly 0 °C, the freezing point of water
- C. Much hotter, since greenhouse gases reflect sunlight away
- D. About the same, since greenhouse gases matter only when people add them

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A city that hosts a winter marathon records its coldest January in 20 years. A resident writes to the paper that this disproves global warming. What is the best response?

- A. The resident is right; a record cold month shows the planet is cooling.
- B. One cold month in one city is weather; climate is judged from long-term global averages, which are rising.
- C. Cold months can never happen in a warming world, so the data must be wrong.
- D. The cold month itself proves warming.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Arctic sea ice, tracked by a satellite program that began in 1979, is shrinking. The darker open ocean absorbs more sunlight, which warms the region further. What kind of feedback is this?

- A. Not a feedback, but an external forcing
- B. A negative feedback
- C. A positive feedback, meaning it helps the climate
- D. A positive (amplifying) feedback

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which of these is a negative (stabilizing) climate feedback?

- A. Melting snow exposes darker ground.
- B. A warmer surface gives off more infrared energy to space.
- C. Warmer air holds more water vapor, a greenhouse gas.
- D. Thawing permafrost releases methane.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Ice cores drilled by an international team that camped on the ice for three summers show glacial and interglacial periods alternating about every 100,000 years. What paced these cycles?

- A. Changes in Earth's orbit and tilt (Milankovitch cycles), amplified by CO2 feedbacks
- B. Individual volcanic eruptions
- C. The 11-year sunspot cycle
- D. Human greenhouse gas emissions

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Scientists at a research institute that publishes an annual state-of-the-climate report find that the stratosphere is cooling while the lower atmosphere warms. What does this pattern indicate?

- A. Natural ocean cycles alone.
- B. The sun has become brighter.
- C. Greenhouse gases are trapping heat in the lower atmosphere, not a brighter sun.
- D. Climate is not changing, since some layers are cooling.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

If recent warming were caused mainly by a brighter sun, which pattern would scientists expect?

- A. Warming in both the lower atmosphere and the stratosphere, with days warming more than nights.
- B. Nights warming faster than days.
- C. No temperature change in any layer.
- D. A cooling stratosphere above a warming lower atmosphere.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A coastal county planning office that recently hired two engineers asks which changes will raise sea level: melting of floating Arctic sea ice, melting of the Greenland ice sheet, or warming of ocean water.

- A. Only melting ice; warming water never changes volume.
- B. Greenland ice melt and ocean warming; floating sea ice adds almost nothing.
- C. All three equally, since all ice is water.
- D. Arctic sea-ice melt is the largest contributor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Global sea level rose during a period when land-ice melt was small. What else can raise sea level?

- A.** Stronger tides
- B.** Melting of floating sea ice
- C.** More evaporation from the ocean
- D.** Thermal expansion of warming seawater

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A climate center whose supercomputer fills a warehouse runs a model that reproduces 20th-century temperatures when given observed inputs. Its projections for 2100 range from about 1.5 °C to over 4 °C of warming. What mainly explains the range?

- A.** Scientists disagree on whether warming is happening.
- B.** Models cannot reproduce the past, so projections are guesses.
- C.** Different assumptions about future greenhouse gas emissions
- D.** The computer is too large to be precise.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A graduate student whose lab shares a building with a geology museum needs atmospheric CO₂ levels from 800,000 years ago. Which record should she use?

- A.** Tree rings
- B.** Satellite measurements
- C.** The continuous Mauna Loa air record
- D.** Air bubbles trapped in Antarctic ice cores

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A science teacher at a school ~~that installed rooftop solar panels last year~~ explains to students why added carbon dioxide warms Earth. Which explanation is correct?

Target word: greenhouse mechanism **Decides:** CO2 interacts with outgoing infrared, not incoming sunlight.

A	Confusion set	Ozone depletion is a separate issue.
B	Inverted operation	Reverses the direction; CO2 is mostly transparent to sunlight.
C	Correct	That slows heat loss to space.
D	Adjacent category	Describes a real greenhouse, not the radiative effect.

2. Answer A Concept 1 • transfer

If Earth had no greenhouse gases at all, what would its average surface temperature be, compared with about 15 °C today?

Target word: reverse: remove the effect **Decides:** The natural greenhouse effect adds about 33 °C.

A	Correct	Without absorption of outgoing infrared, Earth would be far colder.
B	Confusion set	A familiar number, not the computed value.
C	Inverted operation	Reverses their role.
D	Dropped element	Drops the natural greenhouse effect.

3. Answer B Concept 2

A city ~~that hosts a winter marathon~~ records its coldest January in 20 years. A resident writes to the paper that this disproves global warming. What is the best response?

Target word: weather vs. climate **Decides:** One month in one place is weather.

A	Adjacent category	Treats a local weather event as a climate trend.
B	Correct	Climate uses decades of widespread data.
C	Absolute language	Variability continues on top of the trend.
D	Confusion set	A single event cannot prove or disprove the trend.

4. Answer D Concept 3

Arctic sea ice, ~~tracked by a satellite program that began in 1979~~, is shrinking. The darker open ocean absorbs more sunlight, which warms the region further. What kind of feedback is this?

Target word: classify the feedback **Decides:** The change amplifies itself.

A	Adjacent category	It is a response to warming that loops back, which defines a feedback.
B	Inverted operation	Negative feedbacks dampen change.
C	Confusion set	“Positive” means amplifying, not beneficial.
D	Correct	Less ice → more absorption → more warming → less ice.

5. Answer B Concept 3 • transfer

Which of these is a negative (stabilizing) climate feedback?

Target word: find the dampening feedback **Decides:** A warmer Earth radiates more energy to space.

A	Adjacent category	An albedo feedback, which amplifies.
B	Correct	Increased outgoing radiation pushes back against warming.
C	Inverted operation	That amplifies warming.
D	Confusion set	Also amplifying.

6. Answer A Concept 4

Ice cores drilled by an international team ~~that camped on the ice for three summers~~ show glacial and interglacial periods alternating about every 100,000 years. What paced these cycles?

Target word: cause matched to time scale **Decides:** ~100,000-year cycles match orbital changes.

A	Correct	Orbital cycles run on tens of thousands of years.
B	Adjacent category	Eruptions cool climate for a few years.
C	Wrong stem number	Far too short for 100,000-year cycles.
D	Confusion set	The cycles predate humans.

7. Answer C Concept 5

Scientists at a research institute ~~that publishes an annual state of the climate report~~ find that the stratosphere is cooling while the lower atmosphere warms. What does this pattern indicate?

Target word: attribution fingerprint **Decides:** Warming below with cooling above.

A	Adjacent category	Ocean cycles move heat around but do not cool the stratosphere this way.
B	Inverted operation	A brighter sun would warm the stratosphere too.
C	Correct	Solar warming would heat all layers.
D	Absolute language	Cooling above is part of the greenhouse signal.

8. Answer A Concept 5 • transfer

If recent warming were caused mainly by a brighter sun, which pattern would scientists expect?

Target word: reverse: predict the solar fingerprint **Decides:** More sunlight heats every layer and days most.

A	Correct	Extra sunlight enters from the top and acts during daylight.
B	Confusion set	Also a greenhouse signal.
C	Dropped element	Drops the extra energy input.
D	Inverted operation	That is the greenhouse fingerprint.

9. Answer B Concept 6

A coastal county planning office ~~that recently hired two engineers~~ asks which changes will raise sea level: melting of floating Arctic sea ice, melting of the Greenland ice sheet, or warming of ocean water.

Target word: sea-level contributors **Decides:** Floating ice already displaces its own weight.

A	Absolute language	Thermal expansion is a major contributor.
B	Correct	Land ice adds water; warm water expands.
C	Dropped element	Drops the difference between floating and land ice.
D	Confusion set	Melting floating ice barely changes sea level.

10. Answer D Concept 6 • transfer

Global sea level rose during a period when land-ice melt was small. What else can raise sea level?

Target word: the non-ice contributor **Decides:** Water expands as it warms.

A	Adjacent category	Tides change daily levels, not the global mean.
B	Confusion set	Negligible effect on sea level.
C	Inverted operation	Evaporation removes water from the ocean.
D	Correct	Warmer water takes up more volume.

11. Answer C Concept 7

A climate center ~~whose supercomputer fills a warehouse~~ runs a model that reproduces 20th-century temperatures when given observed inputs. Its projections for 2100 range from about 1.5 °C to over 4 °C of warming. What mainly explains the range?

Target word: source of projection spread **Decides:** Different future emissions pathways.

A	Confusion set	Every scenario warms; the range is about how much.
B	Dropped element	Drops the stated success at reproducing the 20th century.
C	Correct	Scenario choice drives most of the end-of-century spread.
D	Camouflage	Hardware size is irrelevant to the spread.

12. Answer D Concept 8

A graduate student ~~whose lab shares a building with a geology museum~~ needs atmospheric CO₂ levels from 800,000 years ago. Which record should she use?

Target word: record that reaches the time span **Decides:** Only ice-core air bubbles reach 800,000 years.

A	Confusion set	Cover centuries to millennia and record growth conditions, not CO ₂ directly.
B	Adjacent category	Satellites cover only recent decades.
C	Dropped element	Drops the time span; it starts in 1958.
D	Correct	They preserve samples of ancient air.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Greenhouse effect mechanism	Greenhouse gases let most sunlight through but absorb outgoing infrared from the surface and re-emit part of it downward.	Thinking greenhouse gases trap incoming sunlight, or linking warming to the ozone hole.	“Why does CO ₂ warm Earth?”
2	Weather vs. climate	Weather is short-term local conditions; climate is the long-term (about 30-year) average, and warming is judged from global trends.	Using one cold season to judge the climate.	A single cold spell or record month.
3	Feedbacks	A positive feedback amplifies an initial change (ice-albedo, water vapor); a negative feedback dampens it (more outgoing infrared as Earth warms).	Reading “positive” as “good.”	A warming effect that causes more (or less) warming.
4	Natural drivers across time scales	Milankovitch orbital cycles pace glacial-interglacial cycles over tens of thousands of years, amplified by CO ₂ feedbacks; sunspots and volcanoes act over years to decades.	Matching a slow cycle to a fast cause, or the reverse.	A cycle length in the stem.
5	Fingerprints of human-caused warming	Greenhouse warming heats the lower atmosphere while cooling the stratosphere; a brighter sun would warm both.	Attributing the pattern to solar output.	Temperature trends by atmospheric layer.
6	Sources of sea-level rise	Sea level rises from thermal expansion of warming water and from melting land ice; melting floating sea ice adds almost nothing.	Blaming Arctic sea-ice melt for sea-level rise.	Floating vs. land-based ice.
7	Climate model projections	Models are tested by reproducing past climate; spread in end-of-century projections comes mostly from different emissions scenarios.	Reading the projection range as doubt that warming is happening.	A range of projected warming for 2100.
8	Records of past climate	Ice-core air bubbles give CO ₂ back hundreds of thousands of years; tree rings cover centuries to millennia; direct and satellite measurements cover recent decades.	Choosing a record that does not reach far enough back.	A target time span for past data.

Course scope (public description): This course explores the science of climate change and covers how the climate system works; what factors cause climate to change across different time scales and how those factors interact; how climate has changed in the past; how scientists use models, observations, and theory to make predictions about future climate; and the possible consequences of climate change for our planet. The course explores evidence for changes in ocean temperature, sea level, and acidity due to global warming. It covers how climate change today is different from past climate cycles and how satellites and other technologies are revealing

the global signals of a changing climate. Finally, the course looks at the connection between human activity and the current warming trend and considers some of the potential social, economic, and environmental consequences of climate change.